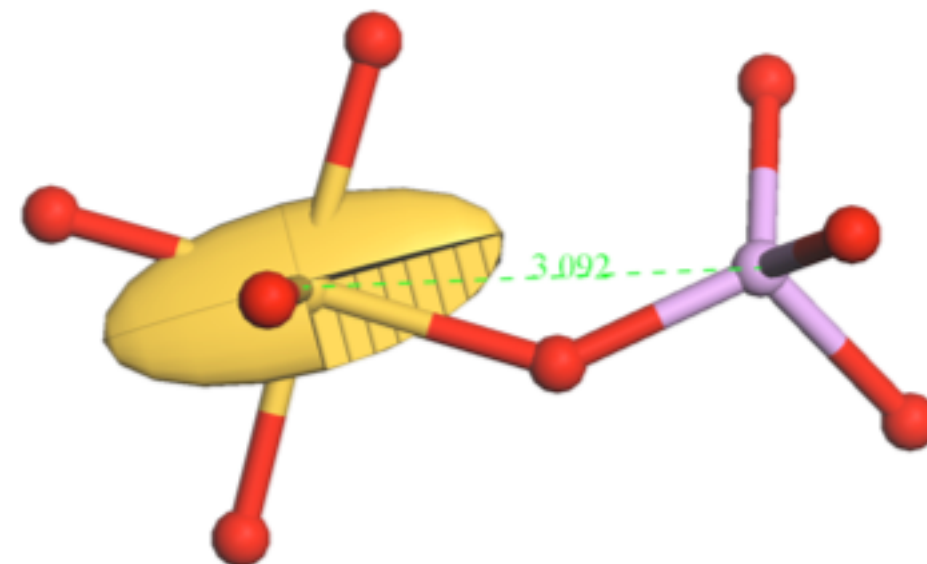
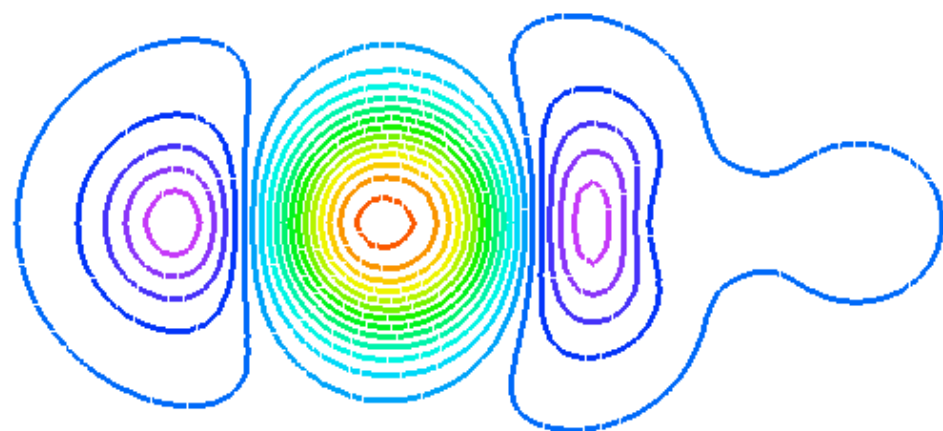


Prediction of NMR parameters in the solid-state

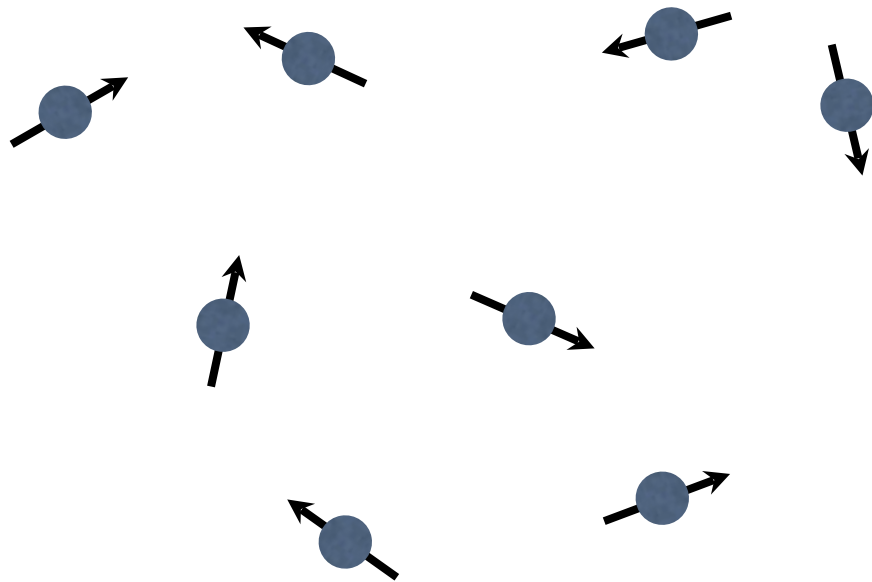
Jonathan Yates

Materials Modelling Laboratory, Oxford Materials

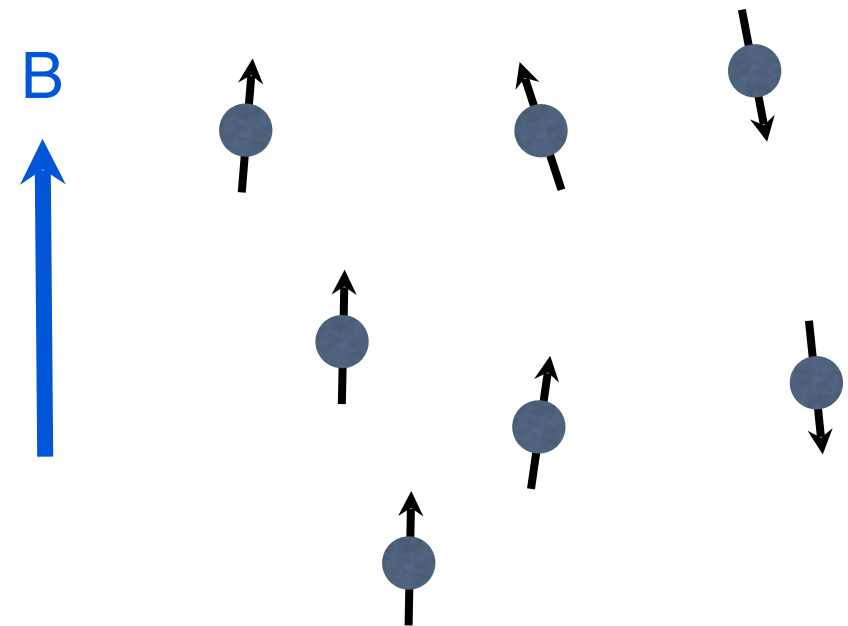


Nuclear Magnetic Resonance

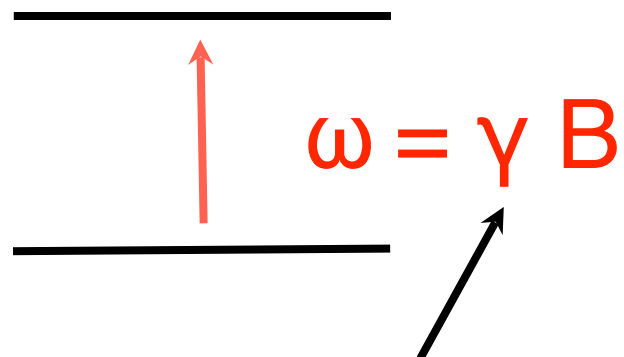
Zero Field



Applied Field



NMR signal is proportional to population difference
For ^1H in 9.4T at 298K



magnetogyric ratio
fundamental nuclear constant

$$\Delta E = 2.65 \times 10^{-25} \text{ J}$$

$$n_{\text{upper}}/n_{\text{lower}} = \exp(-\Delta E / kT) = 0.999935$$

Sensitivity is an issue! Need Large γ and/or B

Common Isotopes

Isotope	Spin	γ $10^7 \text{ T}^{-1} \text{ rad s}^{-1}$	Freq MHz @ 9.4T	Abundance %
^1H	1/2	26.75	400	100
^{13}C	1/2	6.73	100.6	1.1
^{29}Si	1/2	-5.32	79.6	4.7
^{31}P	1/2	-10.84	162.1	100

Common Isotopes

Isotope	Spin	γ $10^7 \text{ T}^{-1} \text{ rad s}^{-1}$	Freq MHz @ 9.4T	Abundance %
^1H	1/2	26.75	400	100
^{13}C	1/2	6.73	100.6	1.1
^{29}Si	1/2	-5.32	79.6	4.7
^{31}P	1/2	-10.84	162.1	100
^{17}O	5/2	-3.63	54.1	0.04
^{49}Ti	7/2	-1.51	22.6	5.41

Solid-State NMR Spectrometers



600 MHz 14.1 T
~£800,000

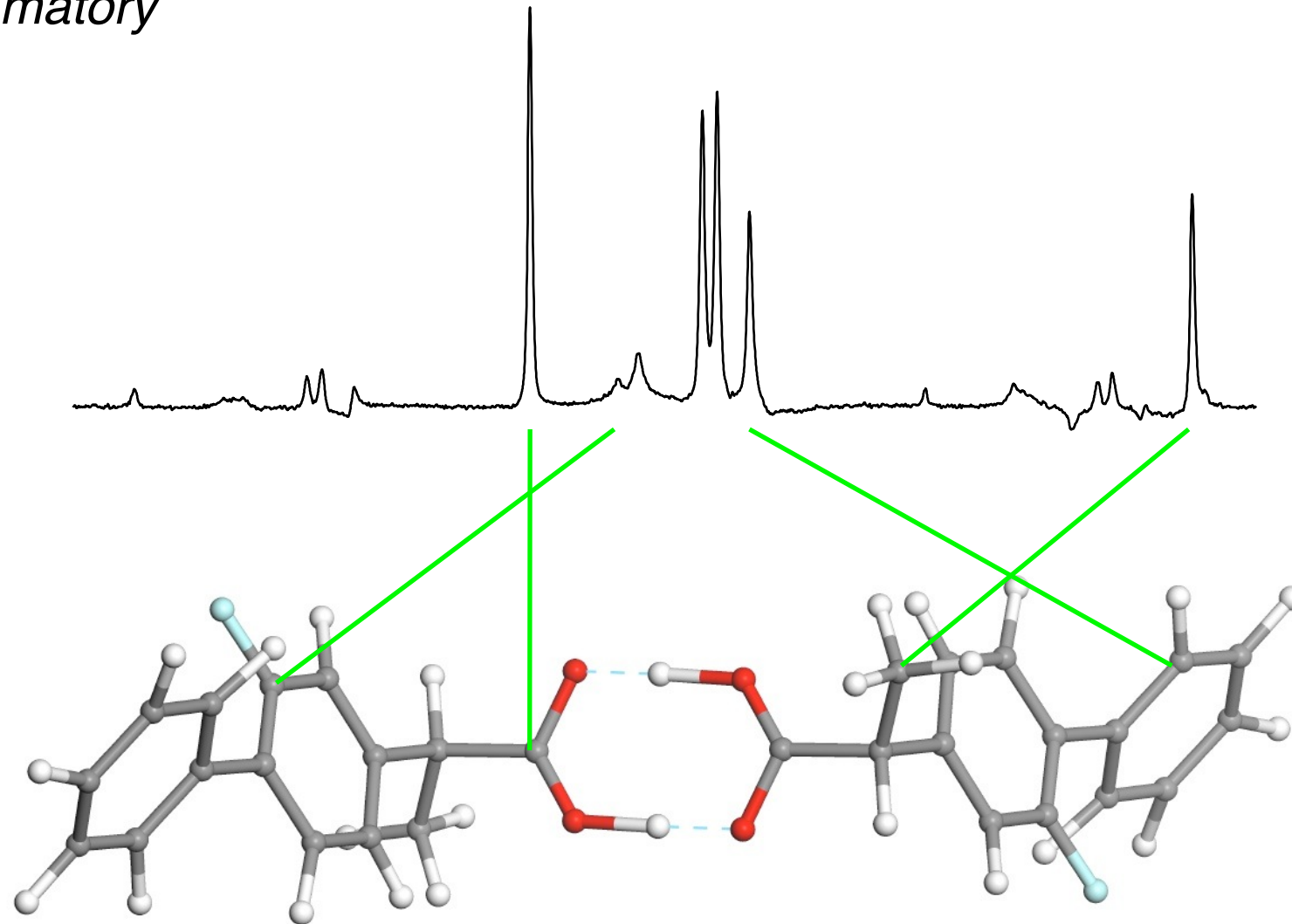
400 MHz 9.4 T
~£300,000

Image courtesy Sharon Ashbrook (St Andrews)

Magnetic Shielding

Flurbiprofen

non-steroidal anti-inflammatory



Each distinct C atom experiences a different magnetic field and resonates at a unique frequency.

Measure the change wrt a standard (for ^{13}C this is liquid tetramethylsilane)

Magnetic Shielding

Measure Larmor frequency

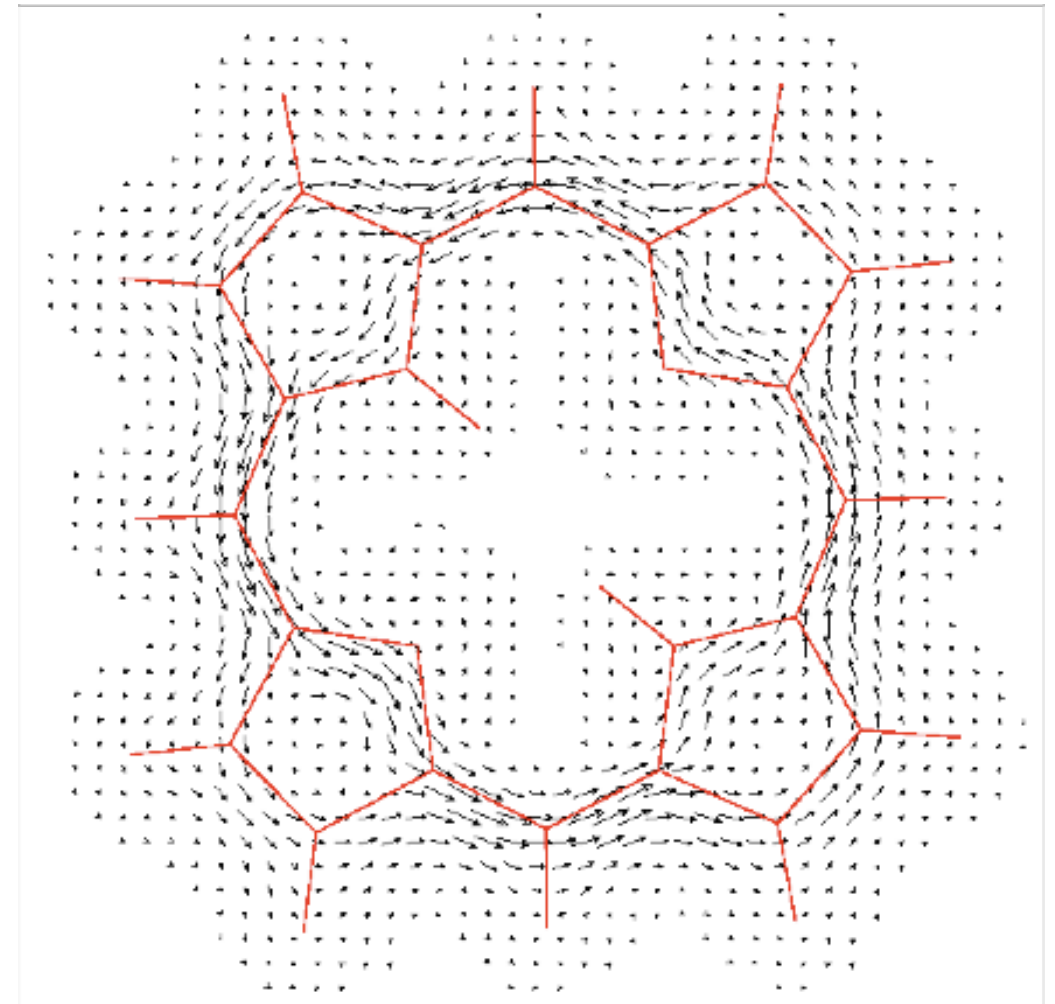
$$\omega = \gamma B_{\text{local}}$$

What is the local field?

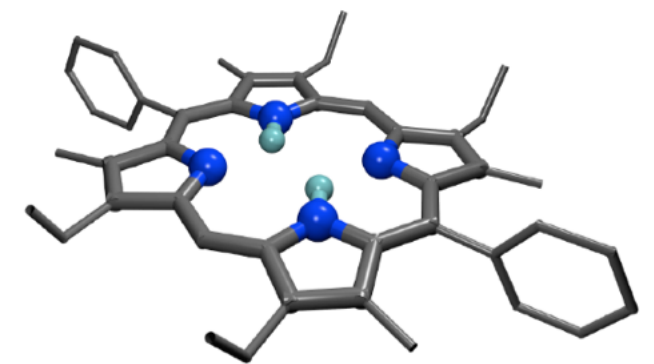
$$B_{\text{local}} = B_0 + B_{\text{induced}}$$

$$B_{\text{local}} = (1 - \sigma) B_0$$

↑
magnetic shielding



Orbital Current induced by B-field in Porphyrin ring



Induced Current

To compute the chemical shifts we just need to calculate the current induced by the external magnetic field

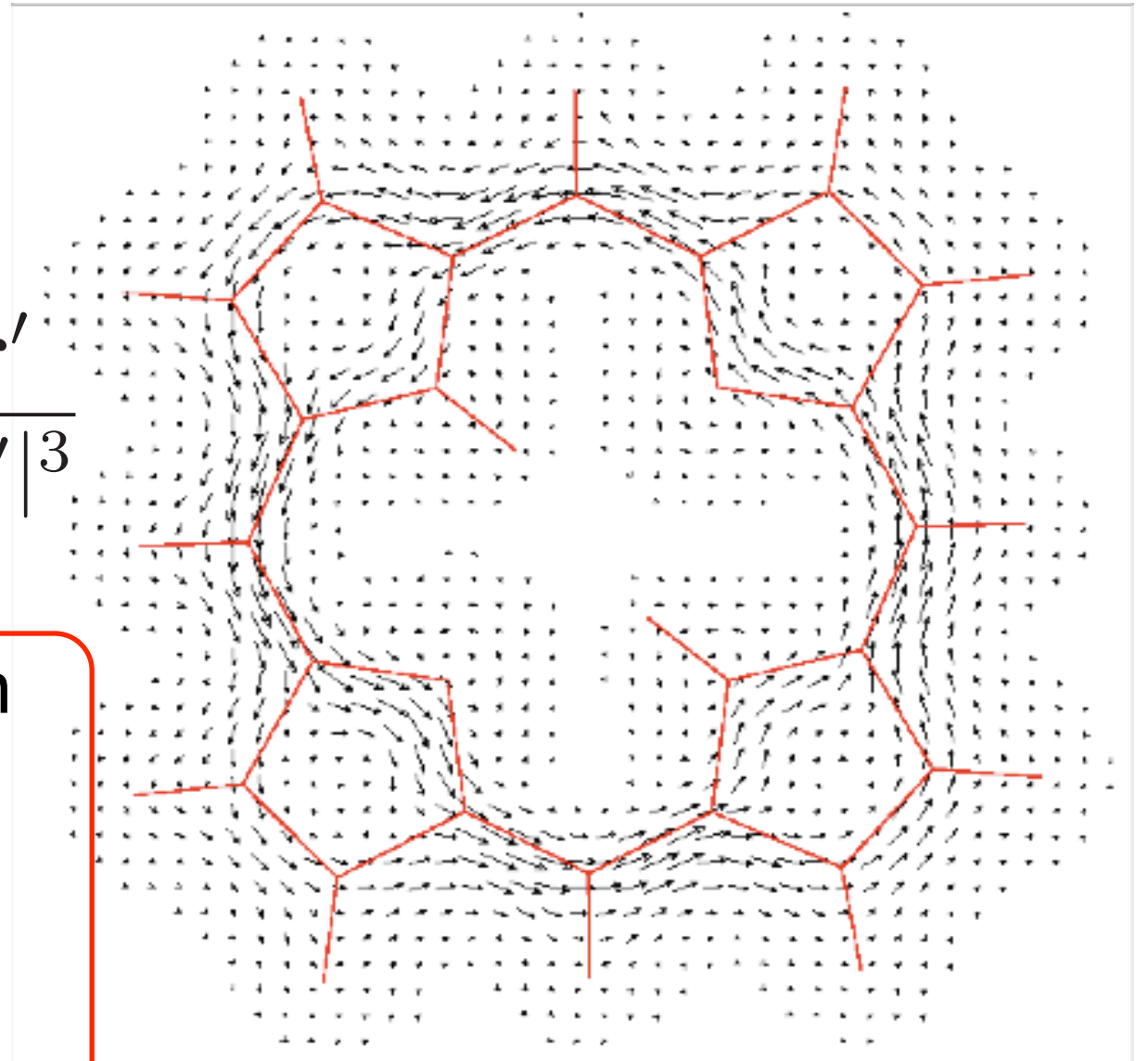
Biot-Savart

$$\mathbf{B}_{\text{in}}(\mathbf{r}) = \frac{1}{c} \int d^3r' \mathbf{j}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}$$

Obtain current within perturbation theory (linear response)

$$O = O^{(0)} + O^{(1)} + \mathcal{O}(\mathbf{B}^2)$$

spin contribution averages to zero in a diamagnetic insulator



$$\mathbf{B} = (1 - \sigma) \mathbf{B}_0$$

$$\mathbf{B}_{\text{in}} = -\sigma \mathbf{B}_0$$

note: σ is a rank 2 tensor

First-principles NMR

Crystal Structure
X-ray, Neutron diffraction
Cambridge Structural Database

C1	-0.602	-0.537	6.384
C2	0.135	0.622	5.689
H1	0.257	0.361	4.639

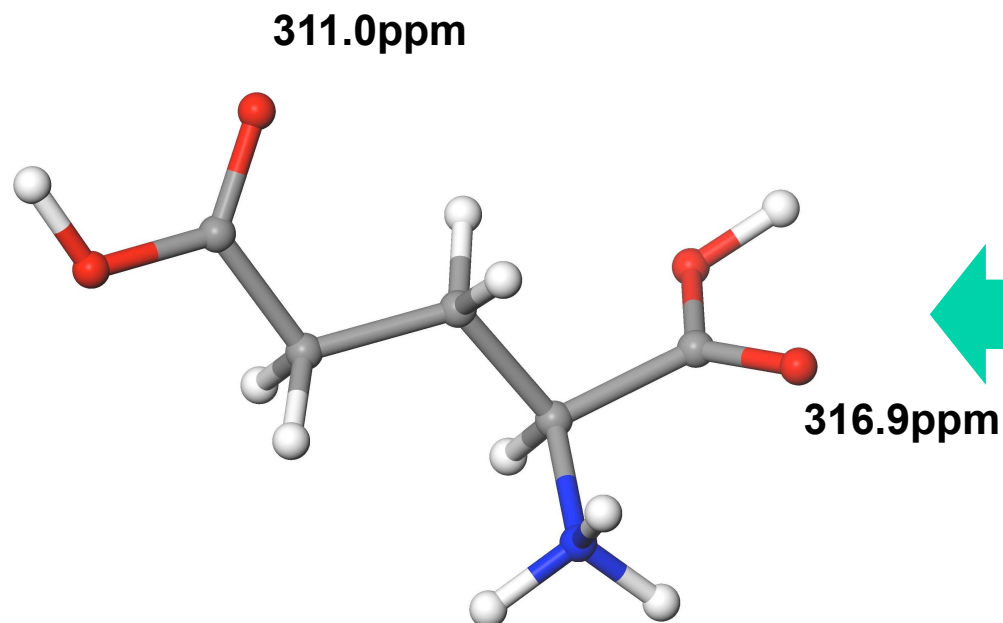
...



Calculate forces
refine structure



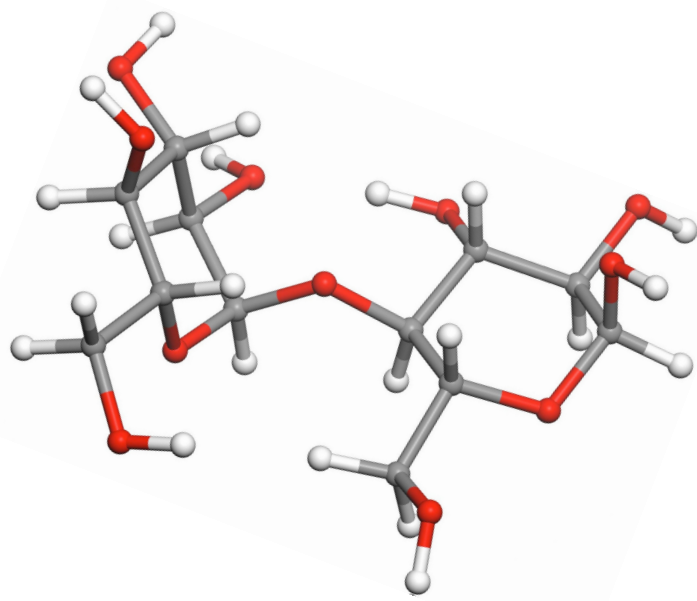
Calculate chemical
shifts



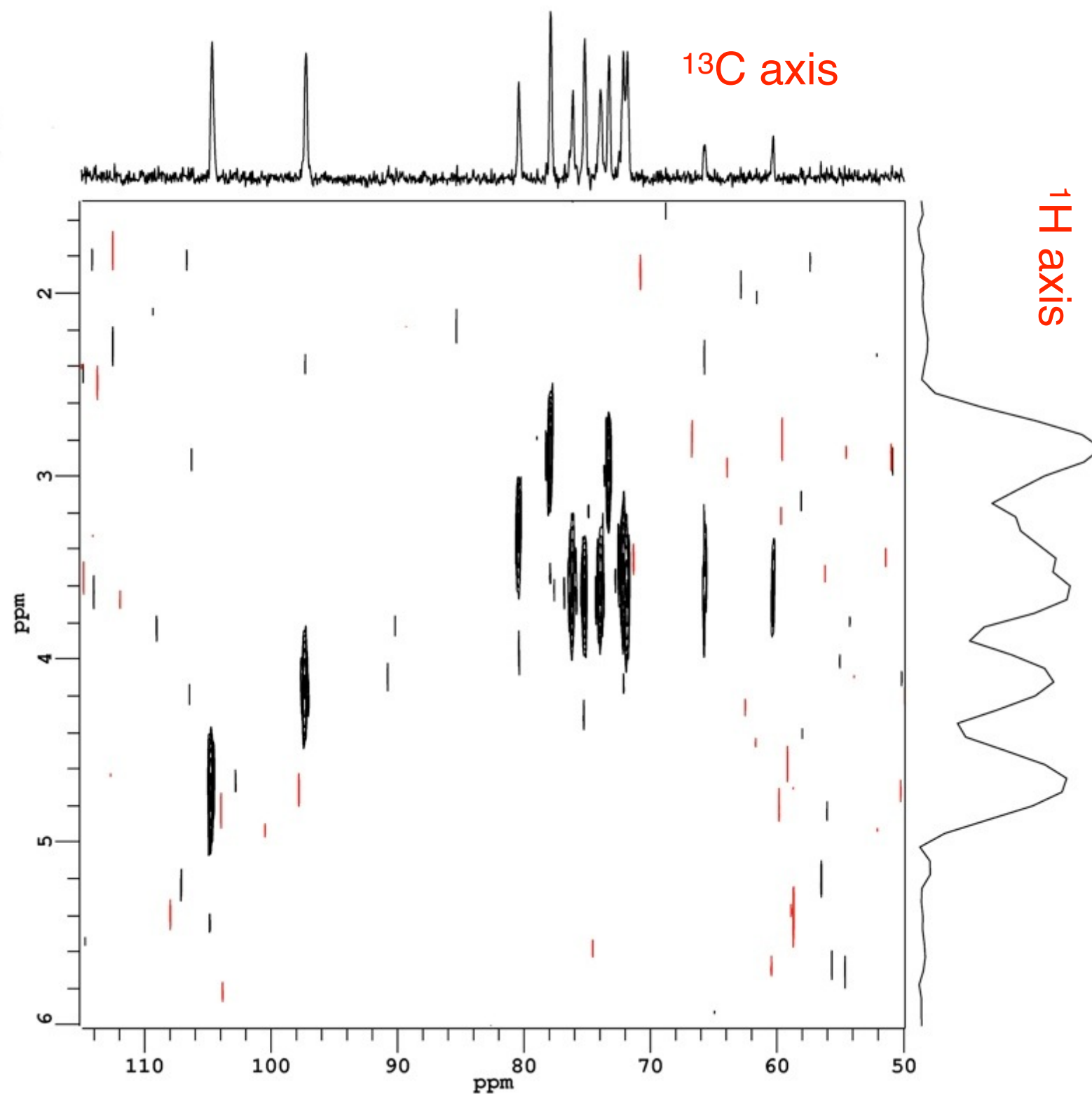
Examples

Maltose

sugar used in brewing



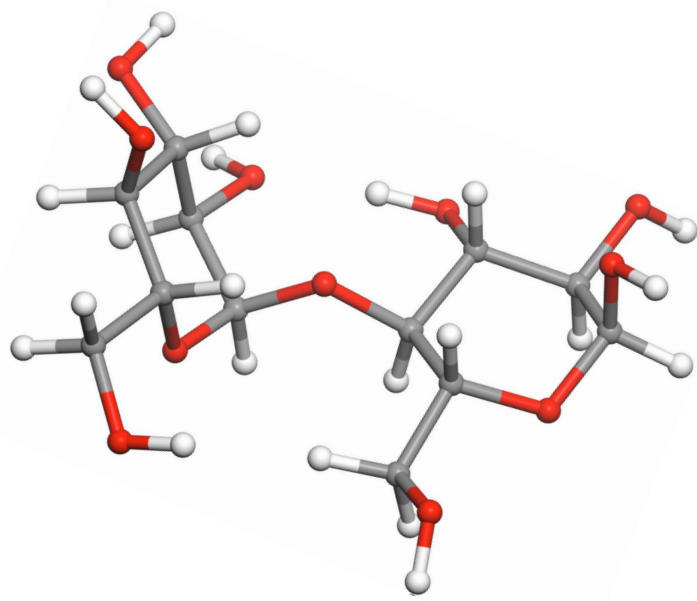
MAS-J-HMQC



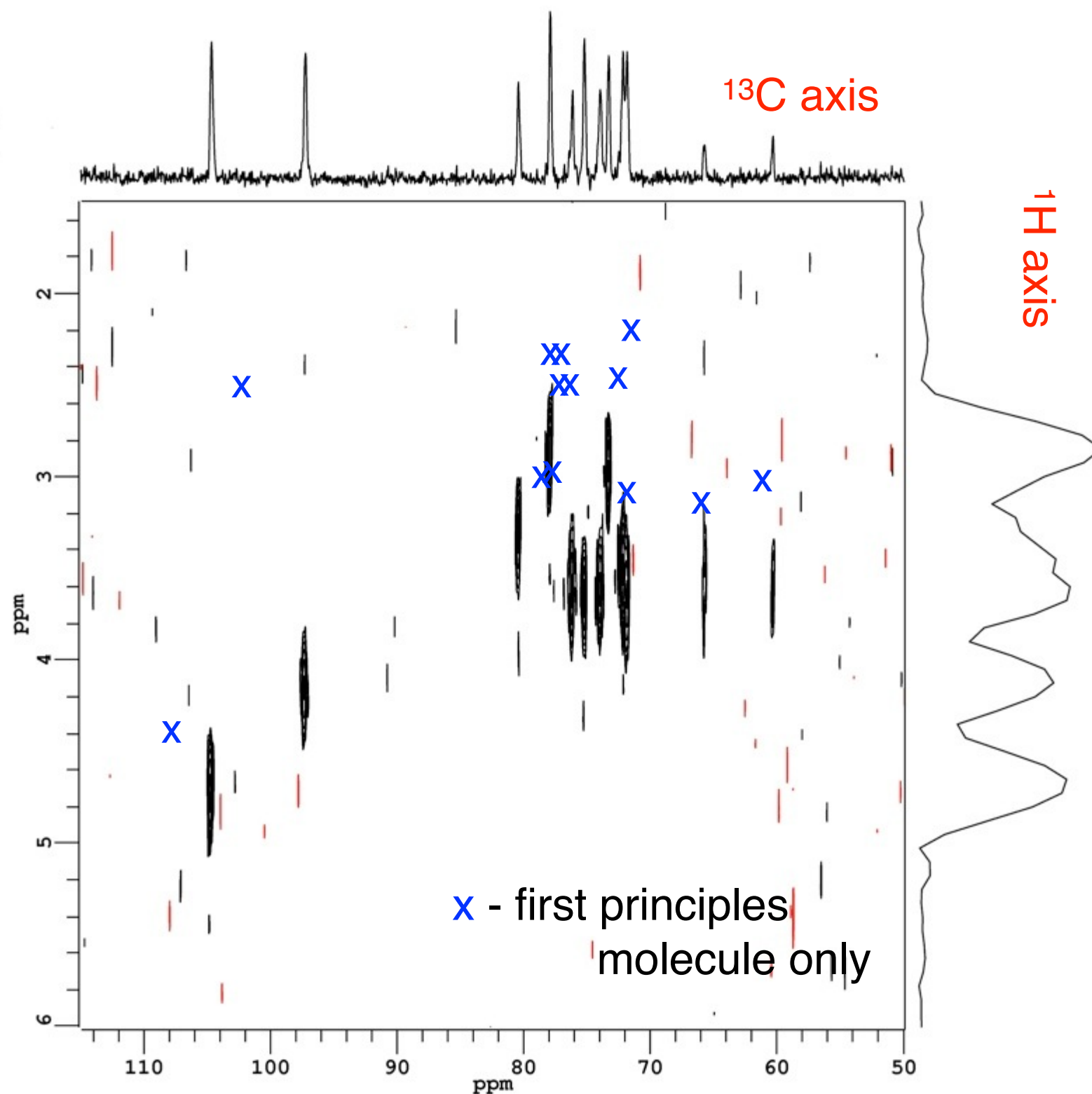
Examples

Maltose

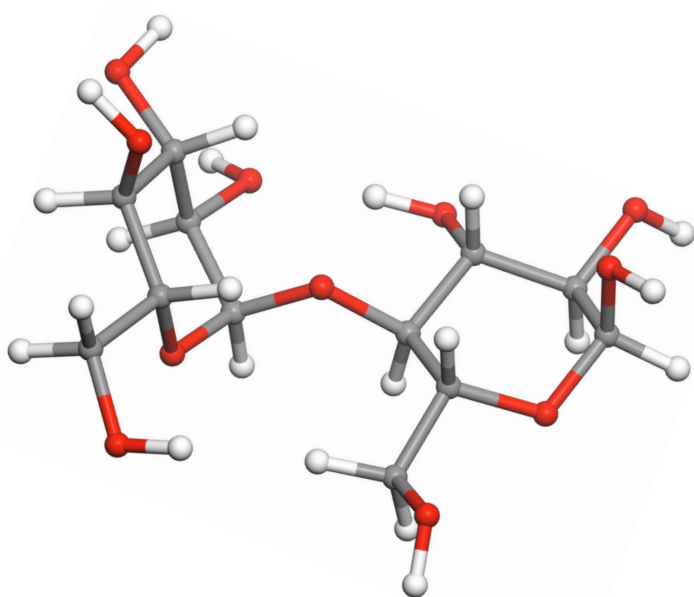
sugar used in brewing



MAS-J-HMQC

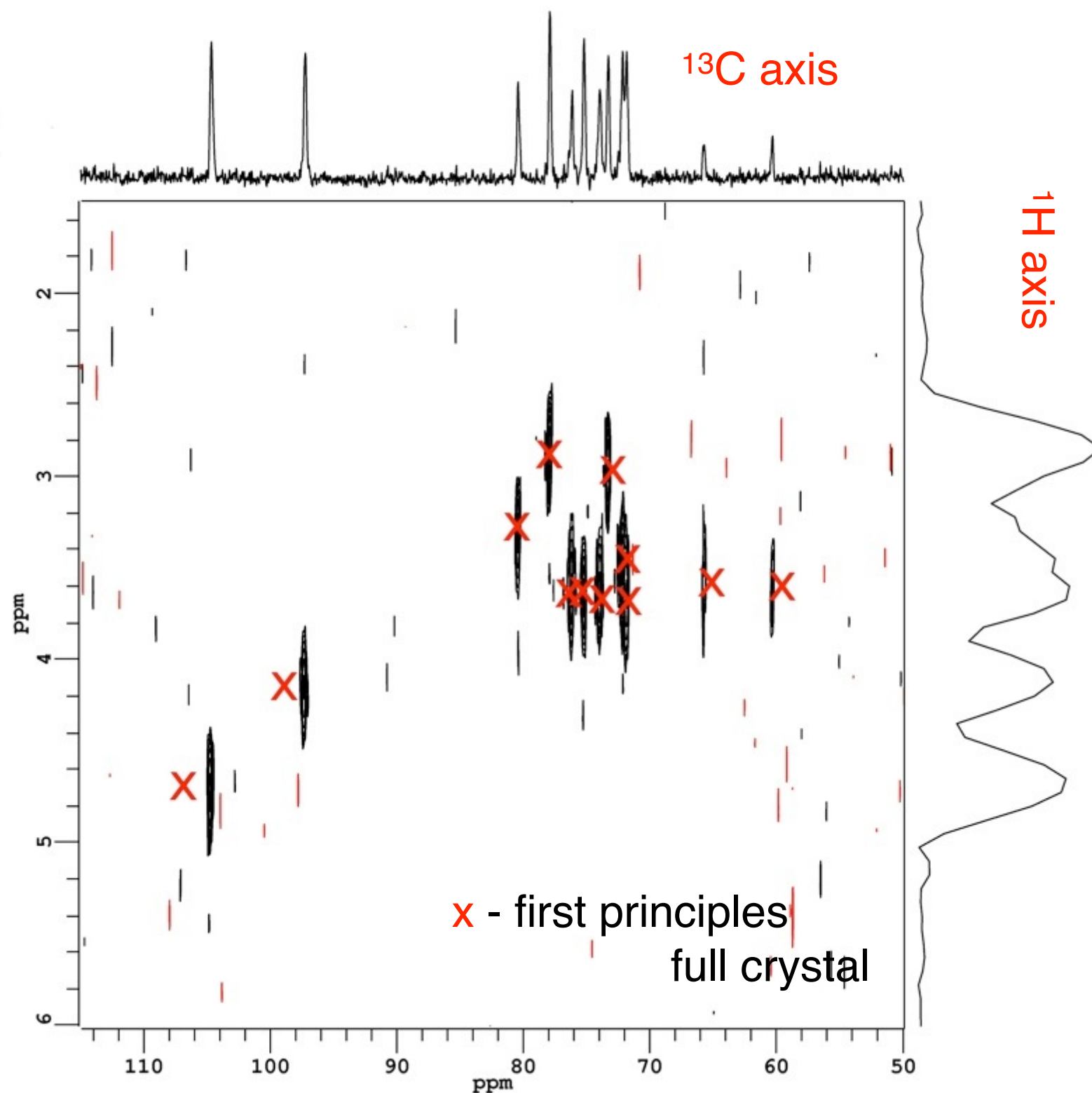


Examples



Molecule to solid
variation due to
intermolecular
interactions (weak
hydrogen bonds)

J. Am. Chem. Soc. 127 10216 (2005)



Magnetic Shielding vs Chemical Shift

Experiment - Chemical Shift

$$\delta_{\text{iso}} = \frac{(\omega - \omega_{\text{ref}}) \times 10^6}{\omega_{\text{ref}}}$$

Result is now independent of the spectrometer frequency and isotope
Standard compounds for reference frequency
beware of those with two standards (eg N)

Calculation - Magnetic Shielding

$$B_{\text{loc}} = (1 - \sigma)B_0$$

σ magnetic shielding

$$\omega = \gamma B_{\text{loc}}$$

Larmor frequency

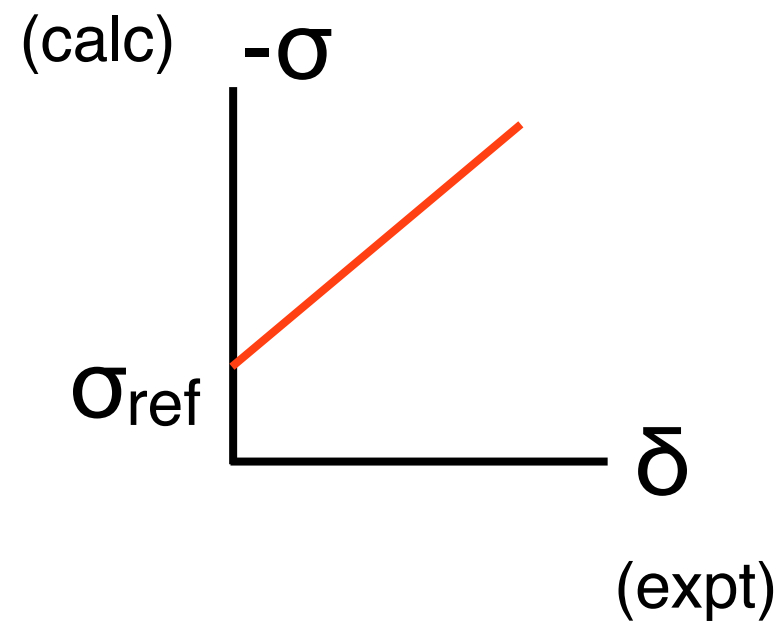
$$\delta = \frac{\sigma_{\text{ref}} - \sigma_{\text{sample}}}{1 - \sigma_{\text{ref}}}$$

$$|\sigma_{\text{ref}}| \ll 1$$

$$\delta = \sigma_{\text{ref}} - \sigma_{\text{sample}}$$

Magnetic Shielding vs Chemical Shift

$$\delta = \sigma_{\text{ref}} - \sigma_{\text{sample}}$$



Compute reference compound
Compute secondary reference
Use literature value
Adjust to match average

Practical Details

*.param file

```
task      : mages  
mages_task : shielding  
           efg  
           nmr  
           jcoupling
```

chemical shift/shielding

electric field gradient

both

Must use on-the-fly pseudopotentials

Highly sensitive to geometry (optimise H X-ray positions)

CONVERGE

(basis cut-off & k-points)

*.castep file

=====					
Chemical Shielding Tensor					

Nucleus		Shielding tensor			
Species	Ion	Iso(ppm)	Aniso(ppm)	Asym	
H	1	23.81	5.27	0.40	
H	2	24.75	-3.35	0.85	
H	3	27.30	-5.79	0.90	
O	5	-43.73	504.95	0.47	
O	6	-63.53	620.75	0.53	
O	7	-43.73	504.95	0.47	
O	8	-63.53	620.75	0.53	
=====					

Anisotropy

$$\sigma_{\text{aniso}} = \sigma_{\text{zz}} - 1/2(\sigma_{\text{xx}} - \sigma_{\text{yy}})$$

Asymmetry

$$\eta = 3(\sigma_{\text{yy}} - \sigma_{\text{xx}})/2\sigma_{\text{aniso}}$$

*.mages file

=====

Atom: O 1

=====

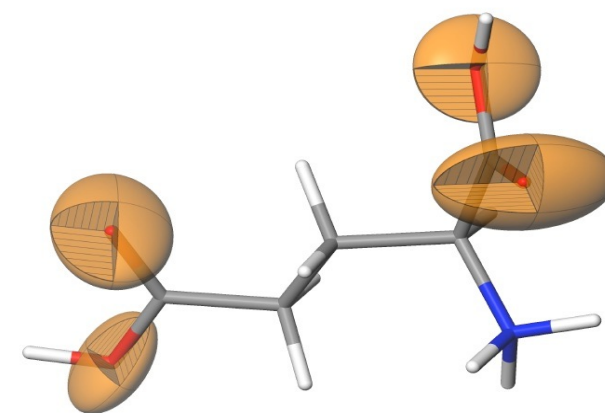
O 1 Coordinates 1.641 1.522 5.785 A

TOTAL Shielding Tensor

218.1858	12.1357	-25.7690
13.4699	191.6972	-7.2419
-25.9178	-6.5205	216.3180

O	1	Eigenvalue	sigma_xx	185.6127	(ppm)	
O	1	Eigenvector	sigma_xx	0.5250	-0.8103	0.2603
O	1	Eigenvalue	sigma_yy	193.8979	(ppm)	
O	1	Eigenvector	sigma_yy	0.4702	0.5310	0.7049
O	1	Eigenvalue	sigma_zz	246.6904	(ppm)	
O	1	Eigenvector	sigma_zz	-0.7094	-0.2477	0.6598
O	1	Isotropic:		208.7337	(ppm)	
O	1	Anisotropy:		56.9351	(ppm)	
O	1	Asymmetry:		0.2183		

Note: shielding tensor has a symmetric and an antisymmetric component. Typical NMR experiments are only sensitive to the symmetric part. Therefore we only diagonalise the symmetric part of the shielding tensor



Magnetic Shielding

$$\sigma_K = \frac{\partial^2 E}{\partial \mathbf{m}_K \partial \mathbf{B}}$$

vector

— $\mathbf{B}_{\text{in}}(\mathbf{r}_K) = -\sigma_K(\mathbf{r})\mathbf{B}_{\text{ext}}$ ← vector

'rank 2 tensor



Magnetic Shielding Tensor

(from a molecular
crystal-calculated)

$$\begin{pmatrix} 82.244 & -38.148 & 49.953 \\ -34.657 & 51.137 & -14.720 \\ 58.131 & 4.405 & 64.137 \end{pmatrix}$$

note: not symmetric
ie $\sigma^T \neq \sigma$

Magnetic Shielding Tensor

(from a molecular
crystal - calculated)

$$\begin{pmatrix} 82.244 & -38.148 & 49.953 \\ -34.657 & 51.137 & -14.720 \\ 58.131 & 4.405 & 64.137 \end{pmatrix}$$

$$\sigma_{sym} = \frac{1}{2} (\sigma + \sigma^T)$$

$$\begin{pmatrix} 82.244 & -36.403 & 54.042 \\ -36.403 & 51.137 & -5.157 \\ 54.042 & -5.157 & 64.137 \end{pmatrix}$$

symmetric

$$M^T = M$$

$$\sigma_{anti} = \frac{1}{2} (\sigma - \sigma^T)$$

$$\begin{pmatrix} 0.000 & -1.746 & -4.089 \\ 1.746 & 0.000 & -9.563 \\ 4.089 & 9.563 & 0.000 \end{pmatrix}$$

antisymmetric

$$M^T = -M$$

$$\begin{pmatrix} 82.244 & -38.148 & 49.953 \\ -34.657 & 51.137 & -14.720 \\ 58.131 & 4.405 & 64.137 \end{pmatrix}$$

9 degrees of freedom
DOF

G_{sym}

$$\begin{pmatrix} 82.244 & -36.403 & 54.042 \\ -36.403 & 51.137 & -5.157 \\ 54.042 & -5.157 & 64.137 \end{pmatrix}$$

G_{anti}

$$\begin{pmatrix} 0.000 & -1.746 & -4.089 \\ 1.746 & 0.000 & -9.563 \\ 4.089 & 9.563 & 0.000 \end{pmatrix}$$

$$\begin{pmatrix} 65.839 & 0.000 & 0.000 \\ 0.000 & 65.839 & 0.000 \\ 0.000 & 0.000 & 65.839 \end{pmatrix} \begin{pmatrix} 16.405 & -36.403 & 54.042 \\ -36.403 & -14.703 & -5.157 \\ 54.042 & -5.157 & -1.702 \end{pmatrix} \begin{pmatrix} 0.000 & -1.746 & -4.089 \\ 1.746 & 0.000 & -9.563 \\ 4.089 & 9.563 & 0.000 \end{pmatrix}$$

isotropic

anisotropic
no trace

antisymmetric

1 DOF

5 DOF

3 DOF

$$\begin{pmatrix} 82.244 & -38.148 & 49.953 \\ -34.657 & 51.137 & -14.720 \\ 58.131 & 4.405 & 64.137 \end{pmatrix}$$

σ_{sym}

$$\begin{pmatrix} 82.244 & -36.403 & 54.042 \\ -36.403 & 51.137 & -5.157 \\ 54.042 & -5.157 & 64.137 \end{pmatrix}$$

σ_{anti}

$$\begin{pmatrix} 0.000 & -1.746 & -4.089 \\ 1.746 & 0.000 & -9.563 \\ 4.089 & 9.563 & 0.000 \end{pmatrix}$$

R

$$\begin{pmatrix} 0.656 & 0.070 & 0.752 \\ 0.479 & -0.808 & -0.343 \\ -0.583 & -0.585 & 0.563 \end{pmatrix}$$

↓ rotate

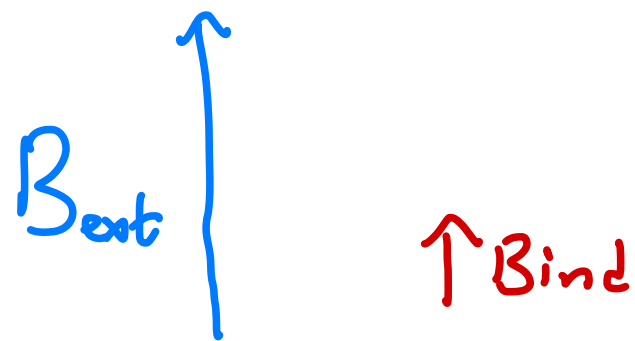
↓ rotate

Principal
Axis
Frame

$$\begin{pmatrix} 7.594 & 0.000 & 0.000 \\ 0.000 & 50.554 & 0.000 \\ 0.000 & 0.000 & 139.369 \end{pmatrix}$$

$$\begin{pmatrix} 0.000 & 9.574 & -2.951 \\ -9.574 & -0.000 & 3.294 \\ 2.951 & -3.294 & 0.000 \end{pmatrix}$$

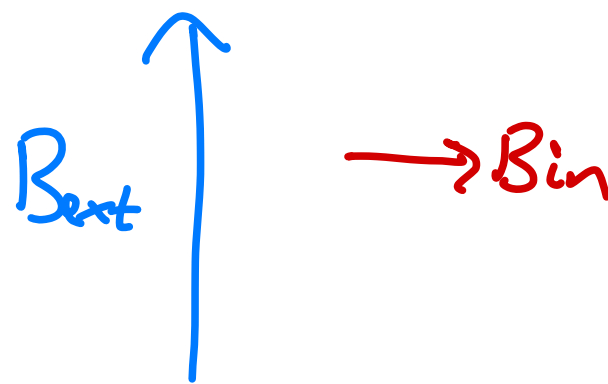
$$\begin{pmatrix} 7.594 & 0.000 & 0.000 \\ 0.000 & 50.554 & 0.000 \\ 0.000 & 0.000 & 139.369 \end{pmatrix}$$



$$B_{Tot} = B_{ext} + B_{ind}$$

1st order effect

$$\begin{pmatrix} 0.000 & 9.574 & -2.951 \\ -9.574 & -0.000 & 3.294 \\ 2.951 & -3.294 & 0.000 \end{pmatrix}$$



$$B_{Tot} = \sqrt{B_{ext}^2 + B_{ind}^2}$$

$$= B_{ext} + \frac{B_{ind}^2}{B_{ext}}$$

2nd order effect

@ 1 GHz 100 ppm chemical shift gives

100 kHz sym

5 Hz anti sym

Note: I'm ignoring the minus sign due to diamagnetism - just to make it easier

NMR Interactions

Chemical Shift

orbital currents create magnetic field

Direct Dipolar Coupling

magnetic field created by neighbouring nuclei

n.b. function of atom positions only i.e. not directly an electronic property

Quadrupolar coupling

nuclei with $I > 1/2$ interact with gradient of electric field (non-spherical charge density)

spin-spin coupling (J-coupling)

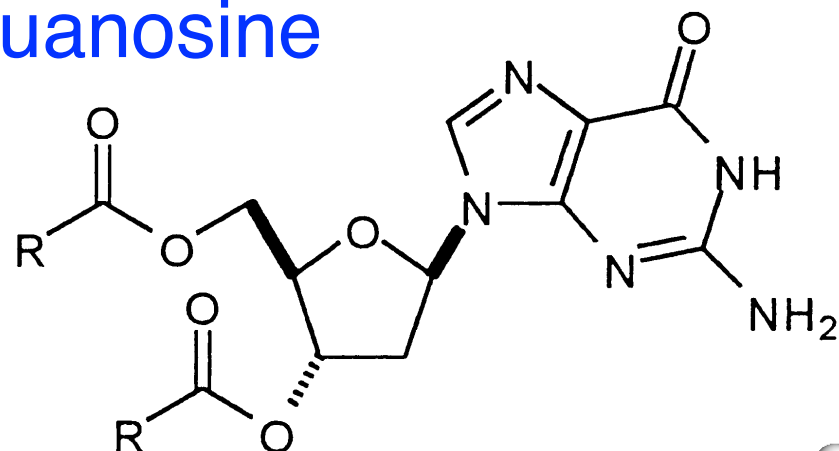
magnetic field induced by neighbouring nuclei but mediated via valence electrons

Simulation of Observed Spectra

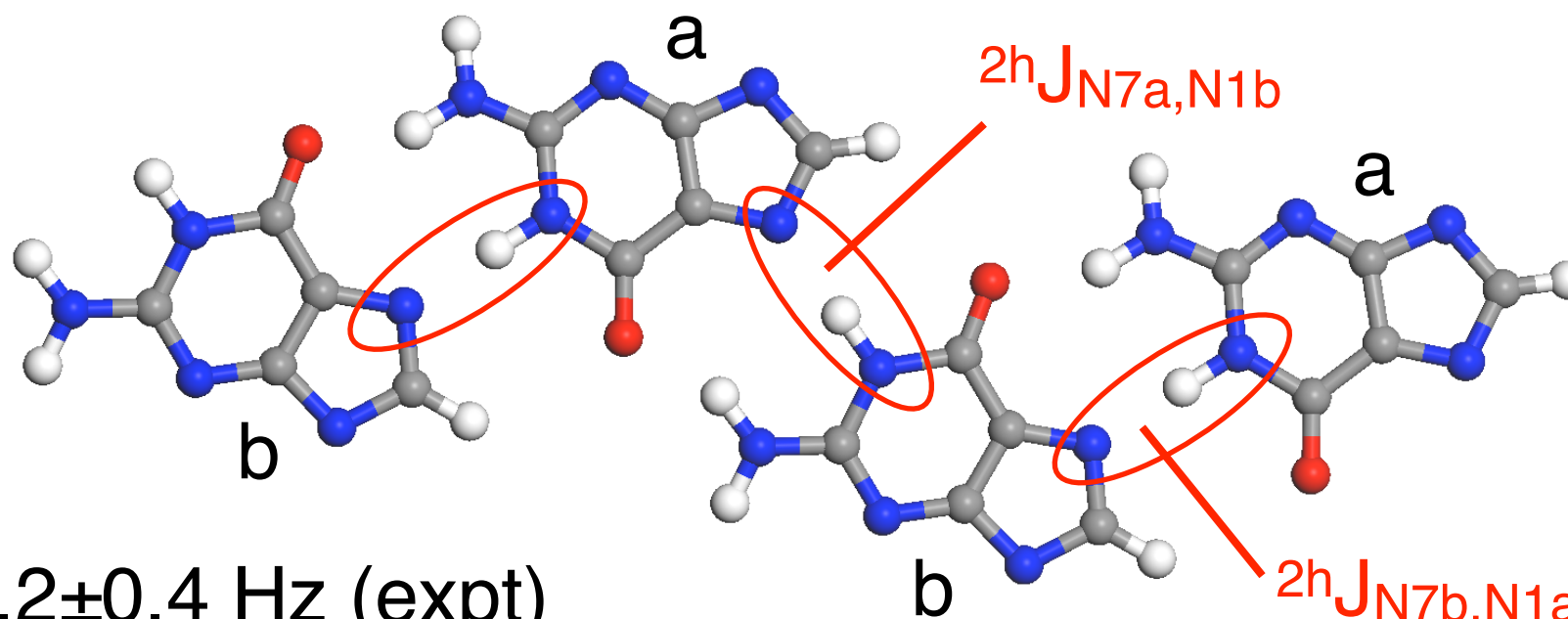
In general not a simple function of NMR interaction tensors. Need to consider experimental conditions (which NMR experiment). Several sophisticated codes to handle this e.g. SIMPSON, Spinevolution (virtual spectrometers)

J-coupling

Guanosine



self-assembles into ribbons
molecular electronics (FET)



$${}^2hJ_{N7b,N1a} = 6.2 \pm 0.4 \text{ Hz (expt)}$$
$$6.5 \text{ Hz (calc)}$$

$${}^2hJ_{N7a,N1b} = 7.4 \pm 0.4 \text{ Hz (expt)}$$
$$7.7 \text{ Hz (calc)}$$

J. Am. Chem. Soc. **130**, 12663 (2008)

Electric Field Gradients

$$E = \int \rho(\mathbf{r}) V(\mathbf{r}) d\mathbf{r}$$

$$E = V_0 \int d\mathbf{r} \rho(\mathbf{r}) V + \sum_{\alpha} \left. \frac{\partial V}{\partial r_{\alpha}} \right|_{r=0} \int d\mathbf{r} r_{\alpha} \rho(\mathbf{r}) + \frac{1}{2} \sum_{\alpha, \beta} \left. \frac{\partial^2 V}{\partial r_{\alpha} \partial r_{\beta}} \right|_{r=0} \int d\mathbf{r} r_{\alpha} r_{\beta} \rho(\mathbf{r}) + \dots$$

Electric Field Gradients

Assume the nucleus is a distribution of charge $\rho(r)$

$$E = \int \rho(\mathbf{r}) V(\mathbf{r}) d\mathbf{r}$$

potential due to electrons & other nuclei

$\frac{\delta E}{\delta r}$ electric field

$$E = V_0 \int d\mathbf{r} \rho(\mathbf{r}) + \sum_{\alpha} \frac{\partial V}{\partial r_{\alpha}} \bigg|_{r=0} \int d\mathbf{r} r_{\alpha} \rho(\mathbf{r}) + \frac{1}{2} \sum_{\alpha, \beta} \frac{\partial^2 V}{\partial r_{\alpha} \partial r_{\beta}} \bigg|_{r=0} \int d\mathbf{r} r_{\alpha} r_{\beta} \rho(\mathbf{r}) + \dots$$

monopole

uniform offset
no effect

dipole

nuclei can't have a dipole moment
so = 0

quadrupole

nuclei with $I > 1/2$ have a quadrupole moment

but not a tensor just a single number, Q

Electric Field Gradients

^{17}O EFG in ethanol
atomic units

$$\begin{pmatrix} 0.1004 & 0.1458 & 1.3802 \\ 0.1458 & 0.0231 & -0.7960 \\ 1.3802 & -0.7960 & -0.1235 \end{pmatrix}$$

symmet
& trace =
5 DoF

$$|V_{zz}| > |V_{yy}| > |V_{xx}|$$

$$V_{xx} = 0.1673 \quad V_{yy} = 1.5138 \quad V_{zz} = -1.6811$$

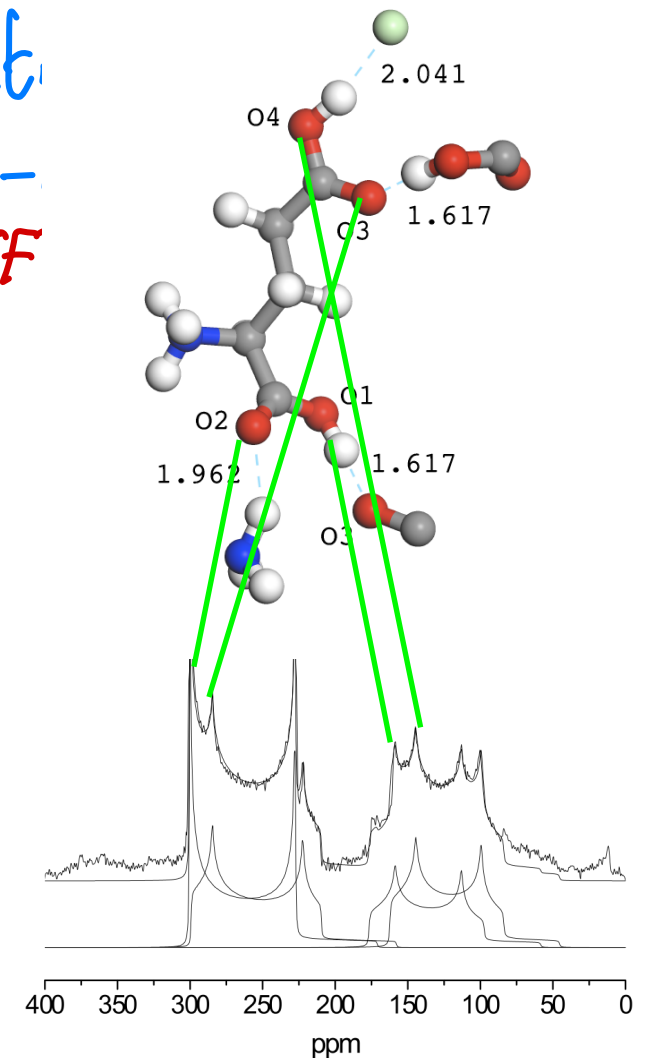
$$C_Q = \frac{eV_{zz}Q}{h}$$

$$\eta_Q = \frac{V_{xx} - V_{yy}}{V_{zz}}$$

$$C_Q = 10.1 \text{ (MHz)}$$

$$\eta_Q = 0.8$$

$$Q(^{17}\text{O}) = -25.58$$



^{17}O MAS Glutamic Acid . HCl

Practical Things

Some practical things to keep in mind

Structure

NMR is very sensitive to structure!

XRD atom positions are often not sufficient for accurate simulation of NMR parameters

Dynamics

Atoms move - spectra are recorded at finite T.

How to account for that in (static) calculations?

Accuracy

How good are our present (approximate) density functionals?

Challenges

Effects of Relativity

scalar relativistic effects easy to treat (relativistic_treatment)
spin-orbit interaction is much harder - but needed for eg shifts on light atoms bonded to heavy atoms

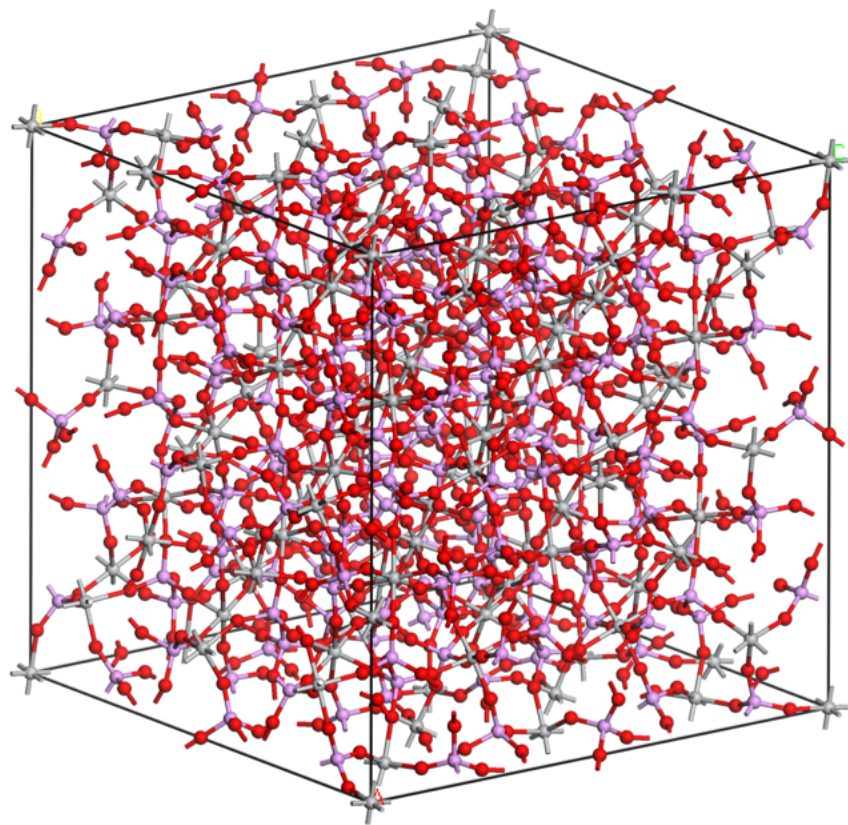
Localised d/f Electrons

Local Density Approximation not enough

Paramagnetic Materials

Magnetic fields from electronic spin
Curie type paramagnetism is hard
Pauli (i.e. Knight Shift) now implemented

Limitations



TiP₂O₇

1020 atoms 8000 electrons

~42 hours on 384 cores (2Tb RAM)

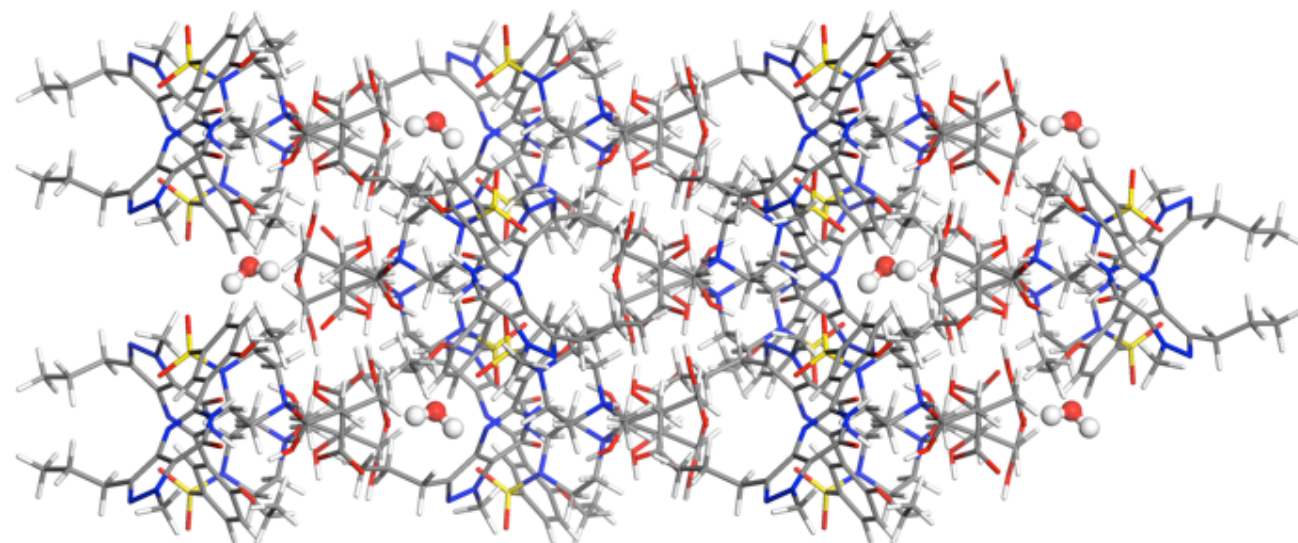
“archer” UK national facility

Pharmaceutical

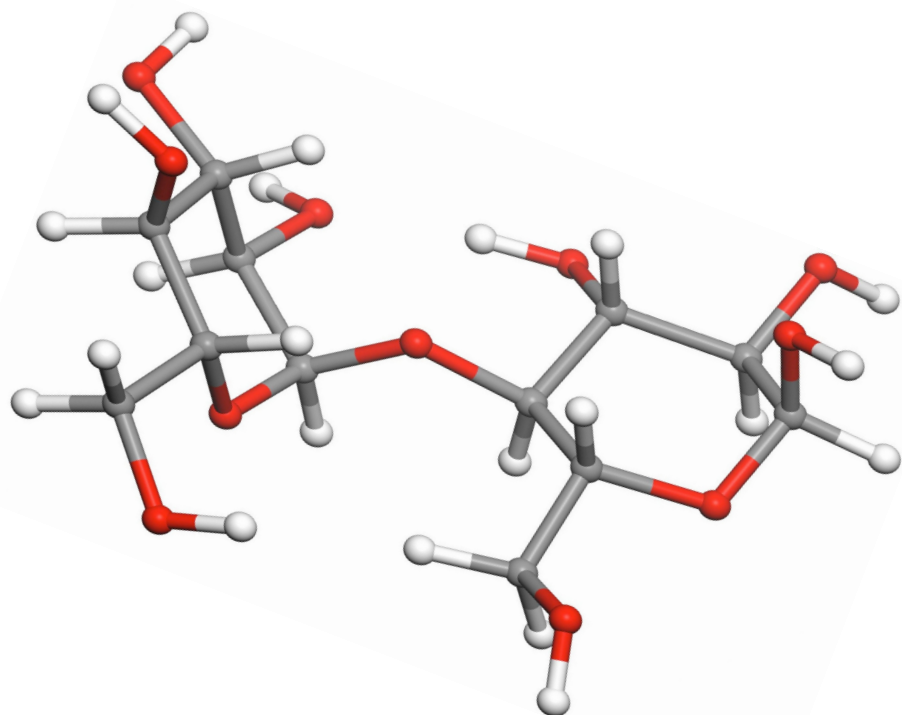
980 atoms 3000 electrons

~24 hours on 96 cores

local cluster



Maltose Again

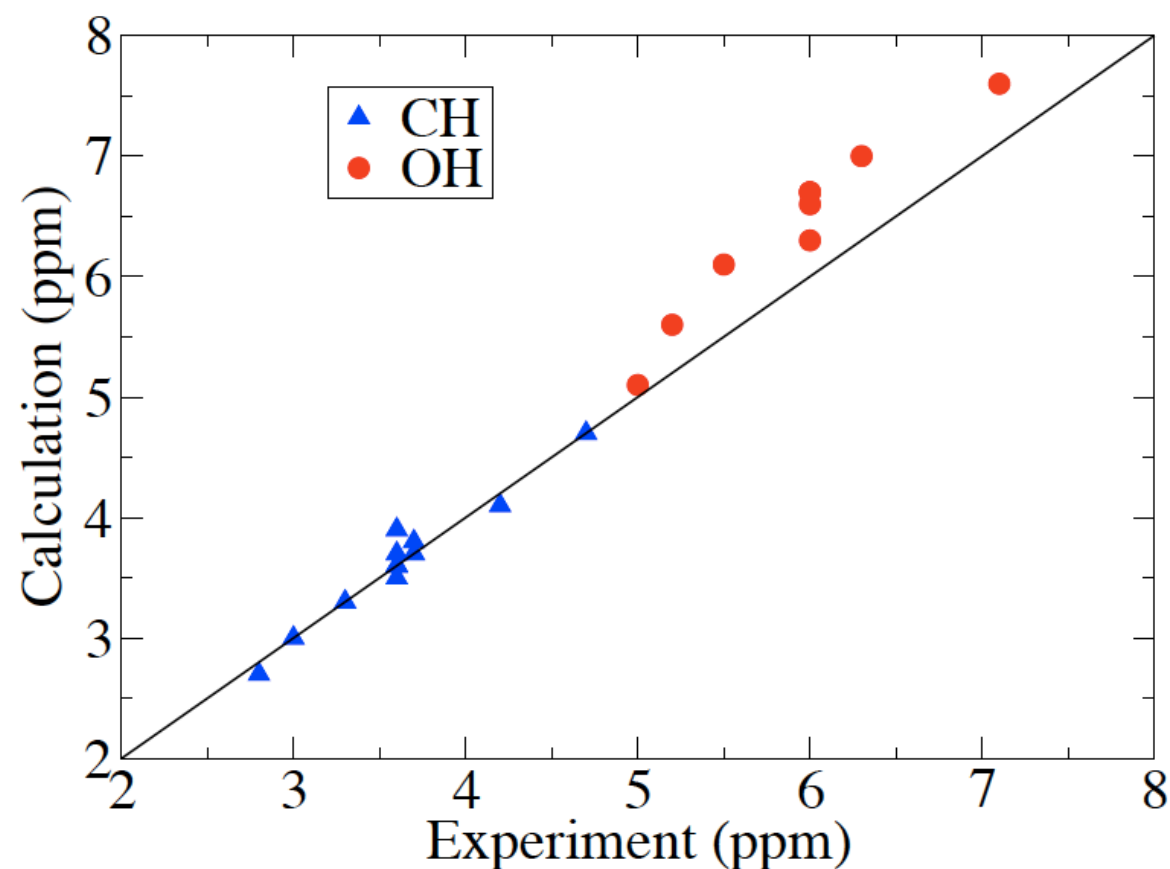


CH

J. Am. Chem. Soc. 127 10216-10220 (2005)

OH

Phys. Chem. Chem. Phys., 12, 6970 (2010)



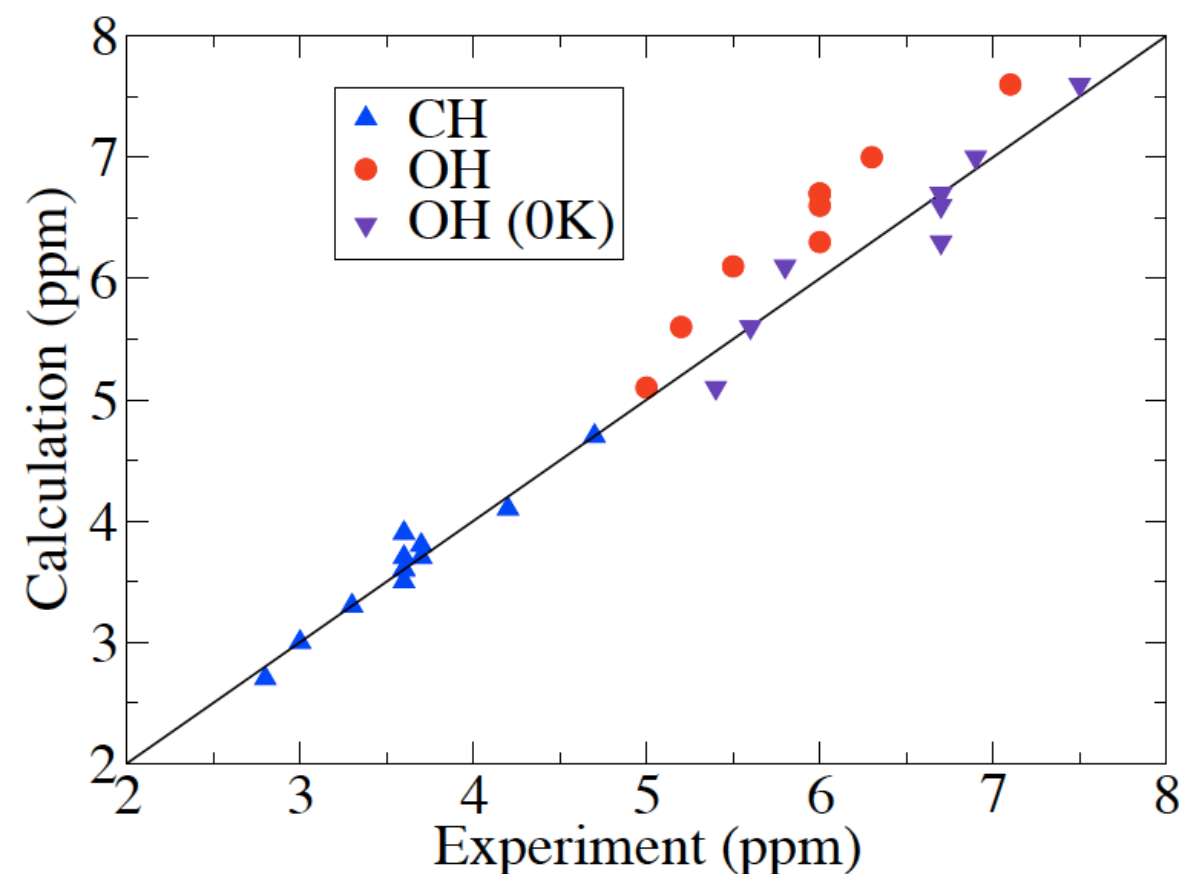
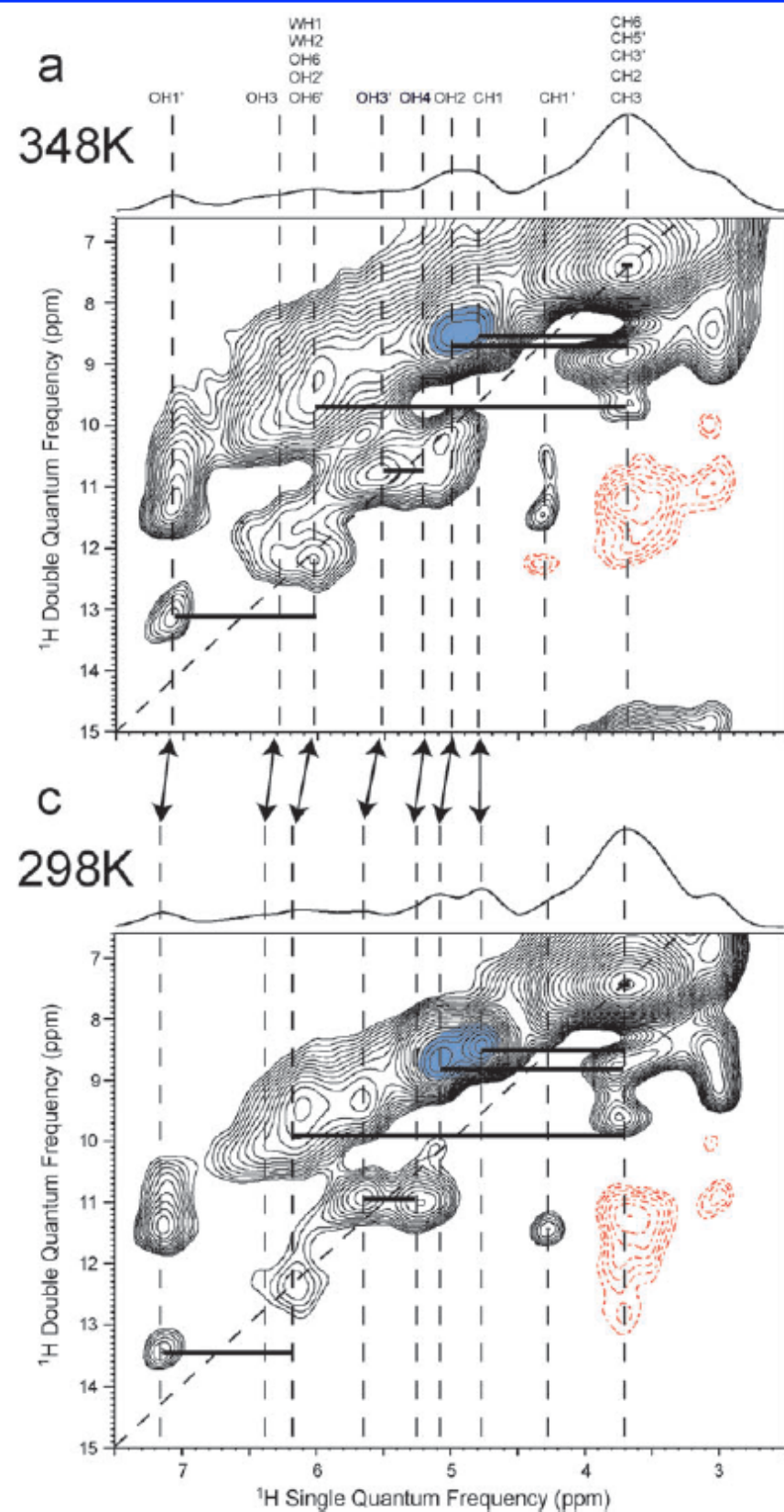
Compare Calculation with experiment

C-H max error 0.3ppm (mean 0.1ppm)

O-H max error 0.6ppm (mean 0.4ppm)

Why are there larger errors for hydroxyl protons?

Variable Temperature



Compare to room temp experiment

C-H max error 0.3ppm (mean 0.1ppm)

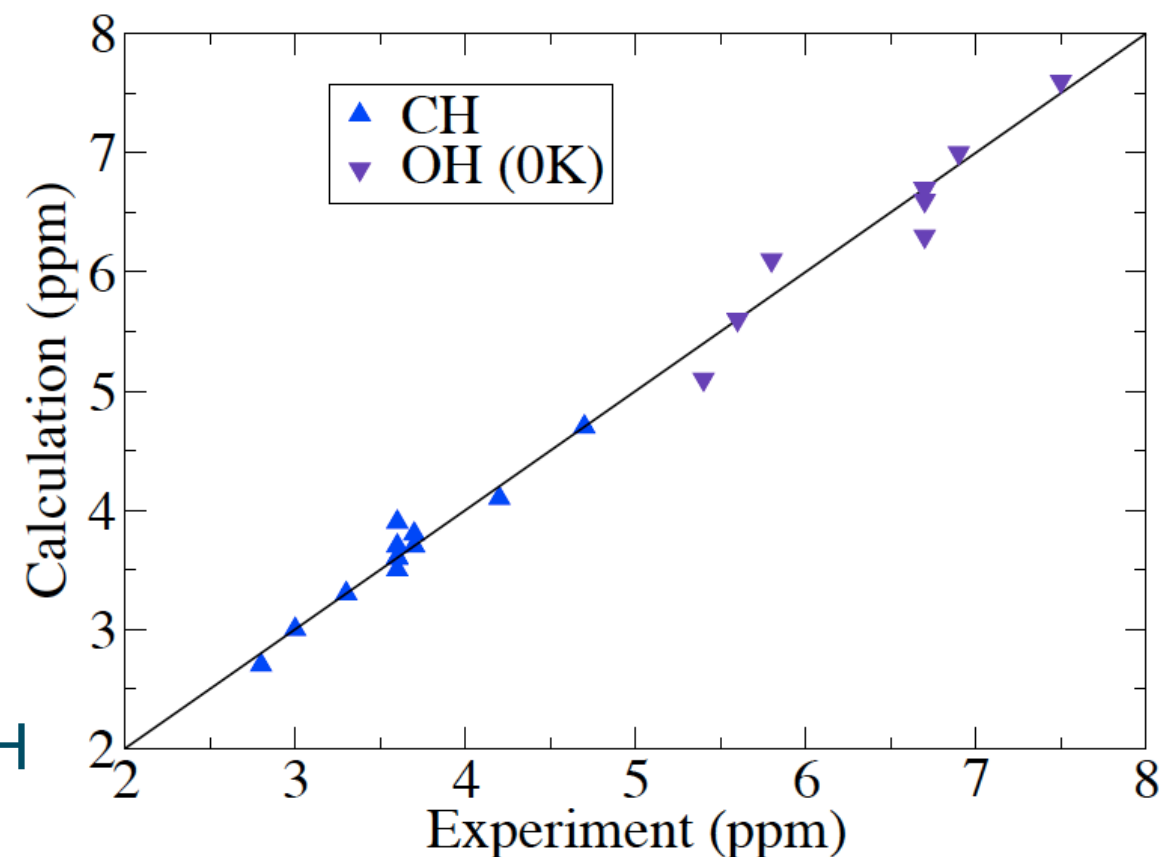
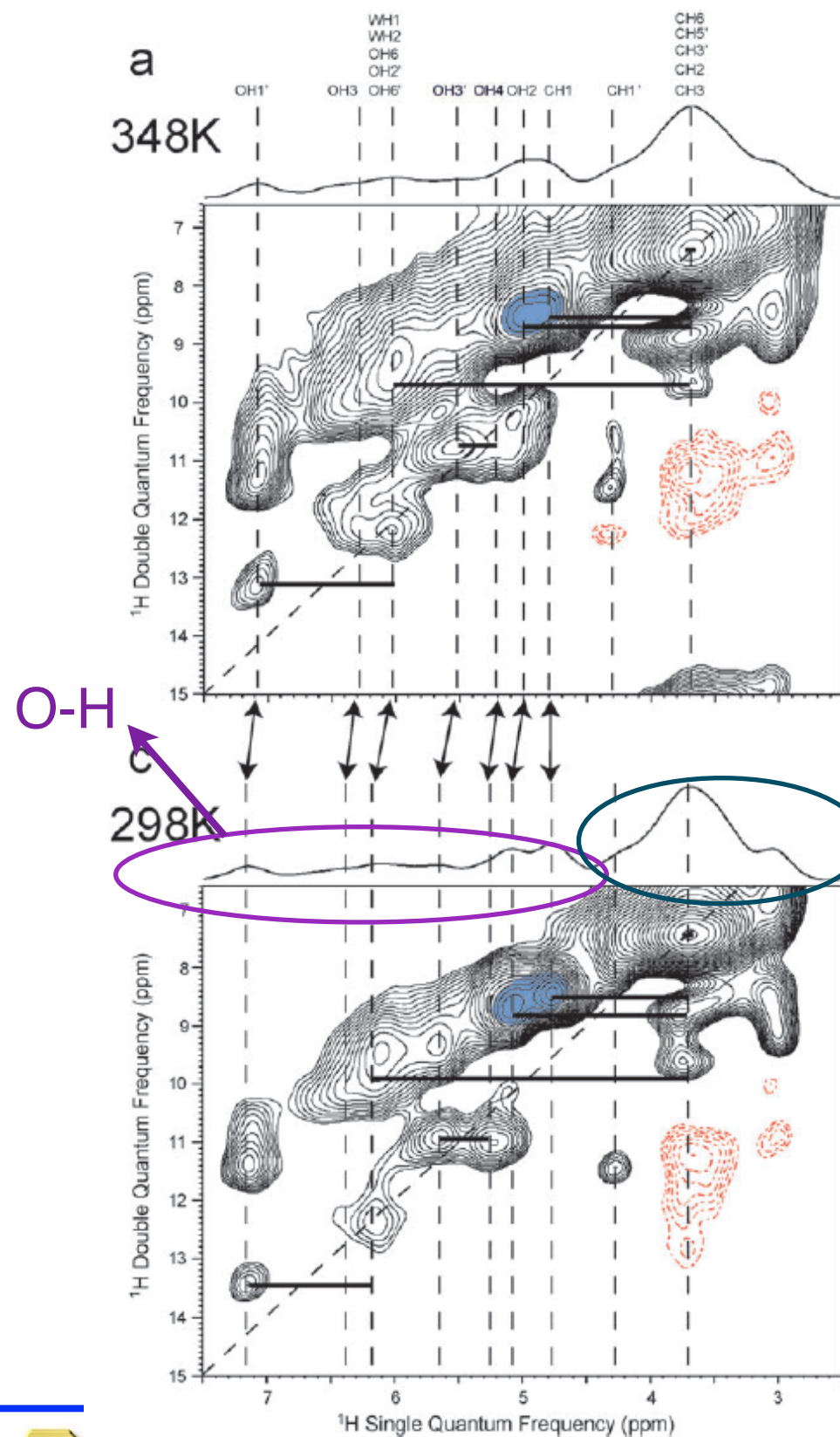
O-H max error 0.6ppm (mean 0.4ppm)

Compare to experiment extrapolated to 0K

C-H (no change)

O-H max error 0.4 ppm (mean 0.1ppm)

Variable Temperature



Compare to room temp experiment

C-H max error 0.3ppm (mean 0.1ppm)

O-H max error 0.6ppm (mean 0.4ppm)

Compare to experiment extrapolated to 0K

C-H (no change)

O-H max error 0.4 ppm (mean 0.1ppm)

Thermal motion

Ways to incorporate thermal effects:

Use a crystal structure obtained at finite temperature

Take into account vibrations:

- Compute the phonon frequencies.

- Generate an ensemble of structures corresponding to a population of phonons.

- Average results.

- Note this gives a correction even at 0K (zero point motion)

Molecular dynamics

see Dumez and Pickard *J. Chem. Phys.* 130, 104701 (2009)

Getting more information

NMR Books

Good Introduction

Nuclear Magnetic Resonance (Oxford Chemistry Primers)

[P. J. Hore](#)

More advanced

Spin Dynamics: Basics of Nuclear Magnetic Resonance

[Malcolm H. Levitt](#)

Solid state NMR

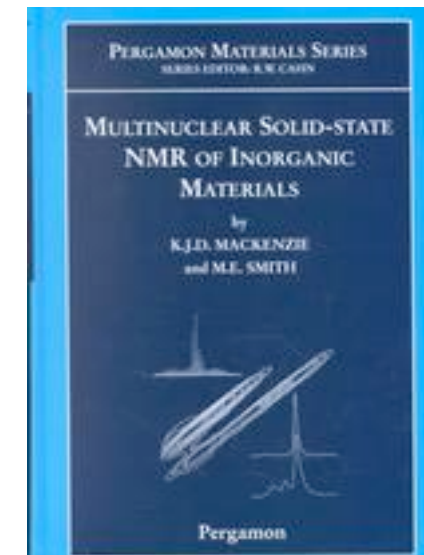
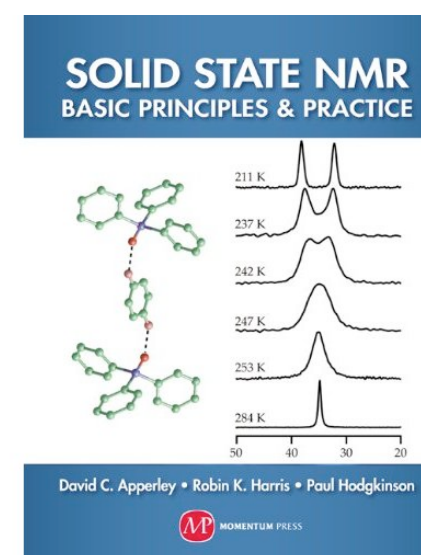
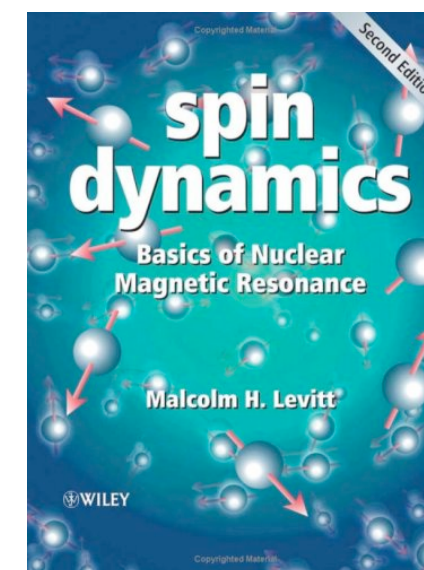
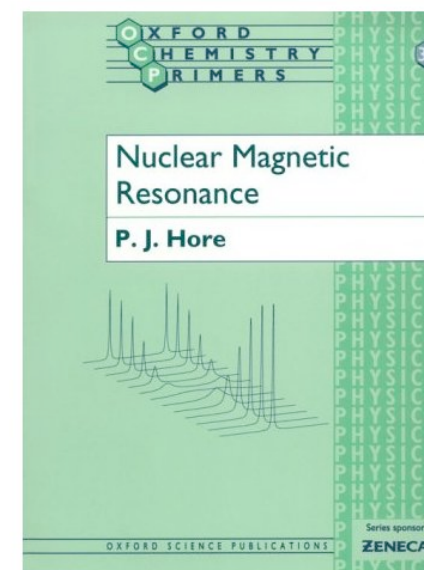
Solid-State NMR: Basic Principles and Practice

David Apperley, Robin Harris, Paul Hodgkinson

Useful survey of applications

Multinuclear Solid-State Nuclear Magnetic Resonance of Inorganic Materials

[Kenneth J.D. MacKenzie](#), [M.E. Smith](#)



Recent Review Articles

Perspective: Current advances in solid-state NMR spectroscopy

J. Chem. Phys. 149, 040901 (2018); <https://doi.org/10.1063/1.5038547>

Sharon E. Ashbrook and Paul Hodgkinson

Getting more information

GIPAW Theory

Exhaustive (and exhausting!) review of applications and theory

First-Principle Calculation of NMR Parameters Using the GIPAW (Gauge Including Projector Augmented Wave) Method: a Chemist's Point of View

Bonhomme, Gervais, Babonneau, Coelho, Pourpoint, Azais, Ashbrook, Griffin, Yates, Mauri, Pickard,
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Calculation of NMR Chemical Shifts for extended systems using Ultrasoft Pseudopotentials

Jonathan R. Yates, Chris J. Pickard, and Francesco Mauri.

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A First Principles Theory of Nuclear Magnetic Resonance J-Coupling in solid-state systems

Sian A. Joyce, Jonathan R. Yates, Chris J. Pickard, Francesco Mauri

J. Chem. Phys. 127, 204107 (2007)

Getting more information

First-principles calculation of NMR parameters

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Abstract

In principle, NMR experiments provide a wealth of information on the structure and dynamics of materials. But how does one access this information? In some cases experience has led to empirical rules which can provide a qualitative interpretation. However, for precise predictions of NMR parameters we must look to first-principles techniques: methods which use only the assumptions of fundamental physics.

